NSE Transaction Cost & Market Impact Simulator

Project Overview | Report generated: 2026-05-19 14:24:10

What This Engine Does

A production-grade Python engine that models every component of trading costs on NSE — from visible charges (brokerage, STT, GST) to the hidden cost that doesn't appear on any contract note: market impact. Built for systematic traders, prop desks, and algo trading firms who need precise cost modelling for strategy evaluation, execution optimisation, and P&L attribution.

Validated against real Dhan broker contract notes with 6/6 component exact match. Market impact powered by live Dhan API data using industry-standard square-root and Almgren-Chriss models.

Why It Matters

Most retail traders estimate costs as 'roughly 0.1% per side.' That's dangerously wrong. A Nifty options trade has a completely different cost profile than an equity delivery trade. An MTF position held for 30 days at ₹25 lakh accumulates ₹8,900+ in interest alone. And a 10,000-share RELIANCE order incurs ~₹2,700 in market impact — yet this number appears nowhere on the contract note.

9 Cost Components Modelled

Component	Charged By	Rate	Notes
Brokerage	Broker	Zero (delivery), ₹20 flat (F&O), min(₹20, 0.03%) intraday	Per broker YAML
STT	Government	Delivery 0.1% (both), Intraday 0.025% (sell), Futures 0.05%, Options 0.15%	Revised Oct 2024
Exchange Charges	NSE/BSE	Equity 0.0031%, Futures 0.0018%, Options 0.0355% (on premium)	Per exchange
SEBI Fee	SEBI	0.0001% (₹10/crore)	All segments
IPFT	Exchange	0.0000001%	Part of GST base
Stamp Duty	Government	Delivery 0.015%, Intraday 0.003%, Futures 0.002%, Options 0.003%	Buy only, round ₹1
GST	Government	18% on (brokerage + exchange + SEBI + IPFT)	NOT on STT/stamp
DP Charges	CDSL/NSDL	₹12.50 + 18% GST per ISIN (Dhan)	Delivery sell only
MTF Interest	Broker	12.49%–16.49% p.a. (5 flat slabs)	Daily, T+1 to settle-1

MTF Interest Slabs (Dhan — Flat, Not Marginal)

Funded Amount	Annual Rate
Up to ₹5,00,000	12.49%
₹5,00,001 – ₹10,00,000	13.49%
₹10,00,001 – ₹25,00,000	14.49%
₹25,00,001 – ₹50,00,000	15.49%
Above ₹50,00,000	16.49%

Flat slab: entire funded amount at the applicable rate (not incremental). Cross-validated: engine ₹102.66 vs Dhan's example ₹102.60.

Market Impact — The Hidden Cost

Market impact is the price displacement your order causes by consuming liquidity. For institutional-size orders, it often dwarfs all regulatory costs combined. It's the difference between the price you expected and the price you actually got.

Square-Root Model (Default)

$$\text{Impact} = \sigma \times \sqrt{(Q / V)} \times \eta$$

Symbol	Meaning	Source
σ	Daily volatility	Std dev of log returns (30-day, Dhan API)
Q	Order quantity	Shares in the order
V	Avg daily volume	30-day mean (Dhan API)
η	Calibration constant	Default 0.3 (configurable in YAML)
Q/V	Participation rate	Fraction of daily volume your order consumes

Key insight: doubling order size increases impact by ~41%, not 100%. The first shares fill at best price; deeper in the book, prices worsen at a decelerating rate.

Almgren-Chriss Model (Advanced)

$$\text{Total Impact} = \text{Temporary Impact} + \text{Permanent Impact}$$

$$\text{Temporary: } \eta \times \sigma \times (Q / (V \times T)) \mid \text{Permanent: } \gamma \times \sigma \times (Q / V)$$

Component	Coefficient	Description
Temporary	η = 0.142	Price displacement that reverts (bid-ask bounce). Proportional to trading rate.
Permanent	γ = 0.314	Information leakage — market learns from your order. Proportional to total qty.
T	Exec. days	Default 1.0. Spreading over days reduces temporary component.

This is what TWAP, VWAP, and IS-optimal execution algorithms use internally. With T=1 (immediate), it gives higher impact than sqrt because it captures both temporary and permanent components. Both models are implemented and switchable via CostEngine(impact_model='almgren_chriss').

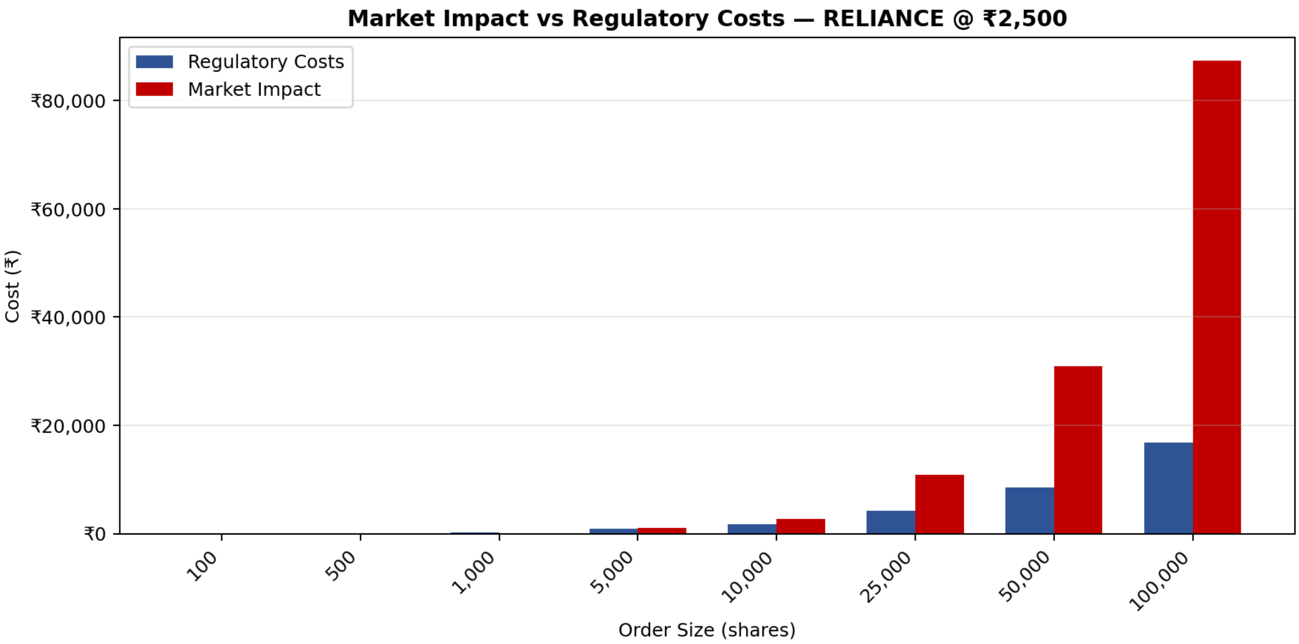
Live Data Integration

Volatility and volume fetched from Dhan's Historical Data API (POST /v2/charts/historical). Current RELIANCE: daily vol 1.69%, avg volume 20,947,697 shares. Credentials read from .env (never committed).

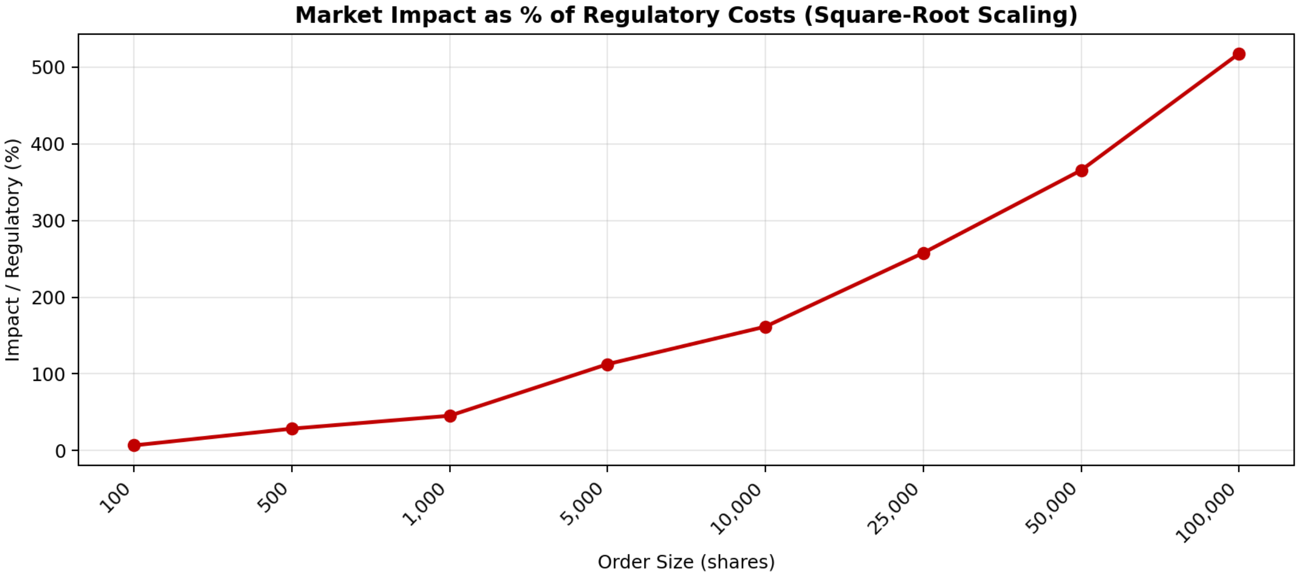
Impact Scenario Analysis

RELIANCE @ ₹2,500 | Daily Vol: 1.69% | Avg Volume: 20,947,697

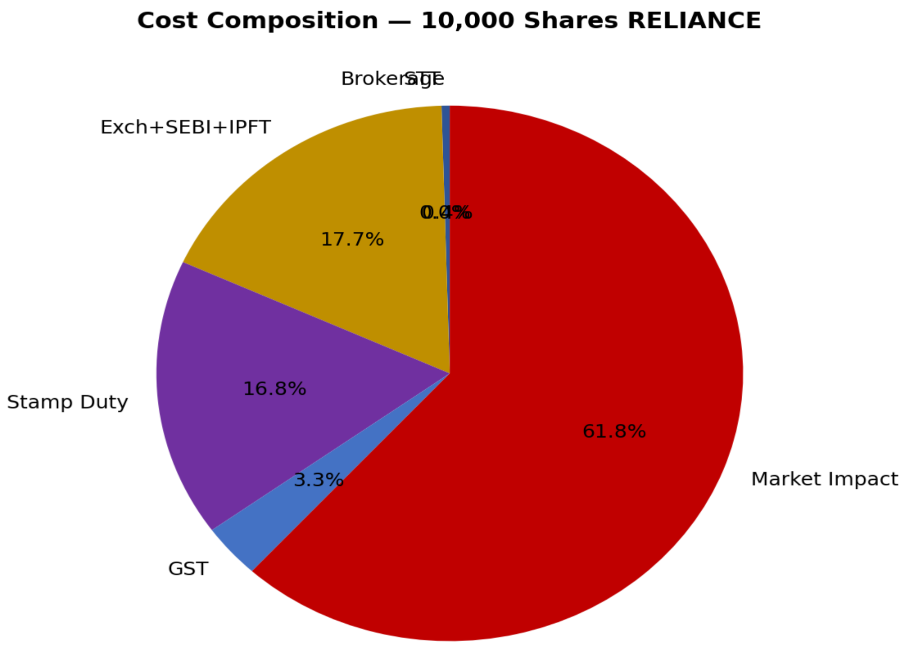
Shares	Trade Value	Regulatory	Impact	Total	Imp/Reg	Imp bps
100	₹250,000	₹40.45	₹2.76	₹43.21	7%	0.1 bps
500	₹1,250,000	₹107.86	₹30.88	₹138.73	29%	0.2 bps
1,000	₹2,500,000	₹192.11	₹87.33	₹279.44	45%	0.3 bps
5,000	₹12,500,000	₹866.17	₹976.37	₹1,842.54	113%	0.8 bps
10,000	₹25,000,000	₹1,708.75	₹2,761.58	₹4,470.33	162%	1.1 bps
25,000	₹62,500,000	₹4,236.48	₹10,916.11	₹15,152.58	258%	1.7 bps
50,000	₹125,000,000	₹8,449.35	₹30,875.41	₹39,324.76	365%	2.5 bps
100,000	₹250,000,000	₹16,875.10	₹87,328.85	₹104,203.95	518%	3.5 bps



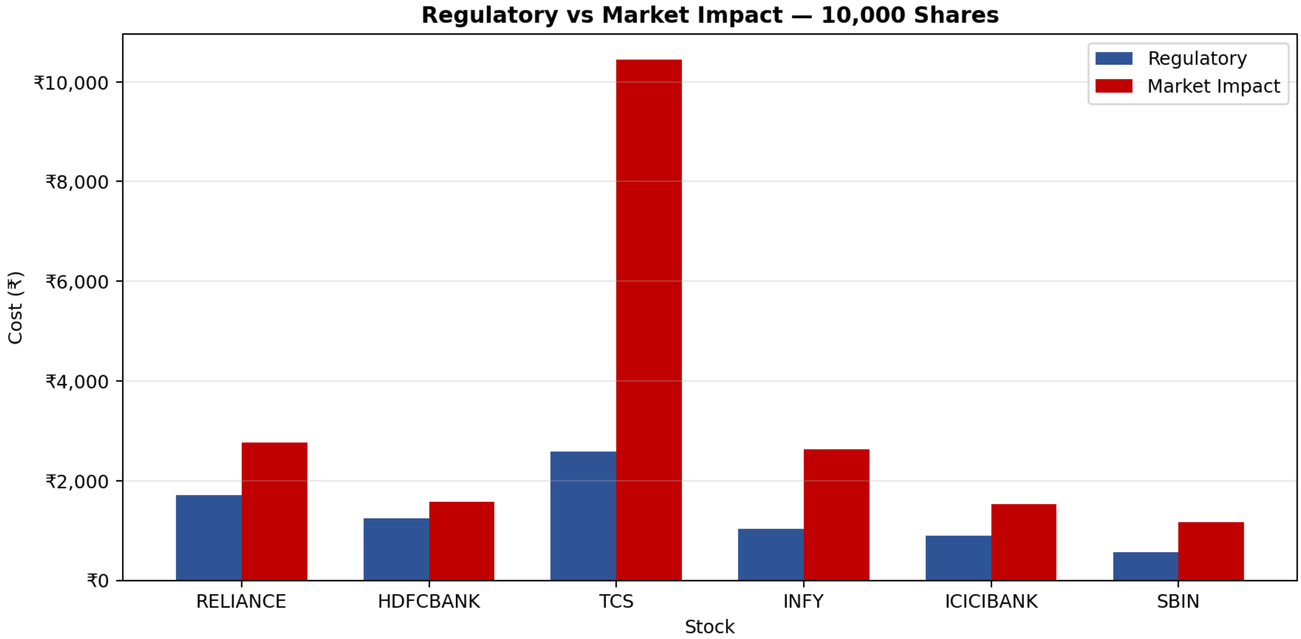
Impact Scaling (Square-Root Curve)



Cost Composition — 10,000 Shares RELIANCE



Cross-Stock Impact Comparison — 10,000 Shares



Stock	Price	Vol (%)	Avg Volume	Regulatory	Impact	Imp/Reg
RELIANCE	₹2,500	1.69%	20,947,697	₹1,708.75	₹2,761.58	162%
HDFCBANK	₹1,800	1.85%	40,126,478	₹1,236.91	₹1,576.54	127%
TCS	₹3,800	1.88%	4,197,124	₹2,585.03	₹10,442.01	404%
INFY	₹1,500	2.17%	13,741,970	₹1,034.69	₹2,630.59	254%
ICICIBANK	₹1,300	1.71%	19,056,514	₹899.88	₹1,529.53	170%
SBIN	₹800	2.15%	19,615,586	₹562.85	₹1,162.94	207%

Contract Note Validation

Validated against a real Dhan contract note dated 08-May-2026 (F&O segment, 2 stock option trades: DELHIVERY 500CE sell, PNBHOUSING 1100CE buy).

Component	Engine	Contract Note	Status
Brokerage	₹40.00	₹40.00	Exact
NSE Transaction Charges	₹12.99	₹12.99	Exact
SEBI Fees	₹0.04	₹0.04	Exact
GST (IGST 18%)	₹9.55	₹9.55	Exact
Stamp Duty	₹1.00	₹1.00	Exact
STT	₹28.00	₹28.00	Exact

Bugs Discovered & Fixed During Validation

1. YAML 'on' keyword: YAML parses 'on:' as boolean True, not 'premium_value'. Would have made options STT 163x too high. Fixed by renaming to 'base_on'.

2. Stamp duty rounding: Brokers round STT and stamp duty to nearest rupee, others to 2 decimals. Added as_contract_note_dict() with correct rules.

3. MTF 5th slab: Original spec had 4 slabs. Dhan confirmed 5th (Above ₹50L at 16.49%). Added to config.

Test Suite: 46 Automated Tests

Covers all segments (equity/F&O/ETF), all trade types (delivery/intraday/MTF), all charge components, slab boundaries, edge cases (penny stocks, ₹10Cr trades), and full integration (round trip, breakeven, what-if).

Architecture

Config-driven: Every rate lives in YAML — zero hardcoded numbers. When SEBI changes rates, update YAML, not Python code.

Modular: 12 independent modules (brokerage, STT, exchange, SEBI, IPFT, stamp duty, GST, DP, MTF, market impact, data feed, engine). Each testable in isolation.

Multi-broker: Broker profiles (Dhan, Zerodha, custom) override defaults. Add a new broker by creating one YAML file.

Module Architecture

Module	Purpose
models.py	Trade, CostBreakdown, RoundTripResult dataclasses
config_loader.py	YAML loading, deep merge, config navigation
brokerage.py	5 models: zero, flat, percentage, min_of, slab
regulatory.py	STT, SEBI fee, stamp duty, IPFT
exchange.py	NSE/BSE transaction charges by segment
tax.py	GST 18% on service charges
dp_charges.py	Depository participant charges + GST
mtf.py	MTF interest — flat and marginal slab modes
market_impact.py	Square-root + Almgren-Chriss models
engine.py	CostEngine orchestrator (calculate, round_trip, breakeven, what_if)
data_feed.py	Dhan API integration for live volatility + volume

Public API

```
engine = CostEngine(broker='dhan')
result = engine.calculate(trade) # Single leg cost
rt = engine.round_trip(buy, sell) # Full round trip with net P&L
be = engine.breakeven(trade) # Min price move to cover costs
alt = engine.what_if(trade, **overrides) # Sensitivity analysis
```

Libraries & Dependencies

Library	Required?	Purpose
pyyaml	Core	YAML config parsing
requests	Optional	Dhan API calls for live vol/volume
python-dotenv	Optional	Load API credentials from .env
openpyxl	Optional	Excel report generation with charts
reportlab	Optional	PDF report generation
matplotlib	Optional	Chart generation for PDF reports

Dhan API Integration

Endpoint	Returns	Used For
POST /v2/charts/historical	Daily OHLCV arrays	Volatility + volume for impact model
Security Master CSV	Symbol to security ID map	images.dhan.co/api-data/api-scrip-master.csv

Project Structure

Path	Description
config/default_rates.yaml	All rates with source comments
config/broker_profiles/dhan.yaml	Dhan overrides + MTF 5-slab config
config/broker_profiles/zerodha.yaml	Zerodha overrides
config/broker_profiles/custom_template.yaml	Template for adding new brokers
nse_cost_engine/ (12 modules)	Core engine package
tests/run_tests.py	46 automated tests
tests/fixtures/contract_note_*.json	Real contract note data for validation
reports/generate_validation.py	Excel report generator (6 sheets, 4 charts)
reports/generate_pdf_overview.py	This PDF report generator
.env.example	API credential template (never commit .env)
requirements.txt / setup.py	Dependencies and pip-installable package

How To Run

Setup:

```
pip install pyyaml
pip install requests python-dotenv openpyxl reportlab matplotlib # optional
```

Run Tests:

```
python tests/run_tests.py # Expected: Ran 46 tests ... OK
```

Configure Dhan API (for market impact):

```
copy .env.example .env # then edit with your credentials
```

Generate Reports:

```
python reports/generate_validation.py # Excel with 6 sheets + 4 charts
python reports/generate_pdf_overview.py # This PDF
```

Future Extensions

Extension	Description
Streamlit Dashboard	Interactive cost calculator with real-time data
Execution Optimiser	Use Almgren-Chriss to find optimal execution schedule for large orders
Historical Cost Attribution	Parse Dhan trade history to compute actual vs estimated costs
Multi-Broker Comparison	Side-by-side cost comparison across brokers for the same trade
Slippage Model	Extend impact with order-book microstructure data (Level 2 / market depth)